

Mappings — project lesson

One hour, worked in pairs: build a mapping, decide whether it is a function, and defend the decision.

LESSON 8

1 × 40 MIN

ALGEBRA 10, §1.1

P1 · 2.1

LESSON CLOCK

5'

8'

20'

10'

2

What a mapping is	5 min
The four kinds	8 min
Pair work — build and classify	20 min
Two pairs present	10 min
Homework	2 min

LEARNING OBJECTIVES

- Describe a mapping by an arrow diagram, a rule and a set of ordered pairs.
- Classify a mapping as one-to-one, many-to-one, one-to-many or many-to-many.
- Decide whether a mapping is a function and justify the decision.
- Present a mapping the pair invented and answer questions about it.

TEXTBOOK

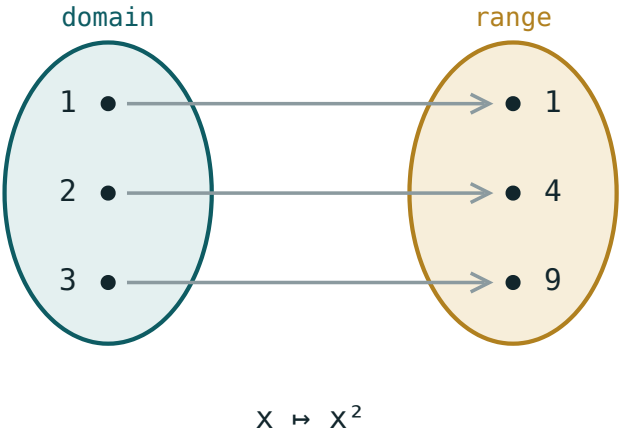
National	pp. 24–27
Cambridge	Pure Mathematics 1, pp. 26–29
Workbook	Project sheet M1

01

Explanation

A mapping is a rule with a direction

Take a set of inputs and a rule. Apply the rule to each input and you get an output. That is a **mapping**. Nothing more is required — the rule does not have to be a formula, and the sets do not have to be numbers.



Four mappings. Read each one left to right and count the arrows leaving, then arriving at, every dot.

ONLY TWO QUESTIONS MATTER

How many arrows leave each input? **How many arrive** at each output? The first answer decides whether the mapping is a function. The second decides whether it can be undone.

The four kinds

KIND	ARROWS OUT OF EACH INPUT	ARROWS INTO AN OUTPUT	EXAMPLE
one-to-one	exactly one	at most one	$x \mapsto 2x + 1$
many-to-one	exactly one	more than one, somewhere	$x \mapsto x^2$
one-to-many	more than one, somewhere	at most one	$x \mapsto \pm\sqrt{x}$
many-to-many	more than one	more than one	student $\mapsto$ subject studied

THE DEFINITION OF A FUNCTION

A mapping is a **function** when every input has **exactly one** output. So the first two rows are functions and the last two are not.

THE COMMONEST ERROR

$x \mapsto x^2$ sends 3 and -3 both to 9. Two arrows arrive at 9 — but only one leaves each input, so it **is** a function. Arrows arriving never break the definition.

The project

In pairs, twenty minutes. Produce one A4 sheet carrying all four of these:

1. A mapping of your own, on a set of at least six inputs, that is **not** numerical.
2. The same mapping as an arrow diagram **and** as a set of ordered pairs.
3. Its classification, with one sentence of justification for each half of the name.
4. One small change to the mapping that turns it into a different kind, and the new name.

MARKING, OUT OF 8

2 for the mapping being genuinely your own · 2 for both representations agreeing · 2 for a correct classification · 2 for the change in part 4 actually working.

Two pairs present in the last ten minutes. The class asks one question each: “*Is it a function, and how do you know from your diagram alone?*”

02 Worked examples

EXAMPLE 1 Classify the mapping $x \mapsto x^3$ on the real numbers.

- 1 Each x has one cube.
One arrow leaves each input.
- 2 Can two inputs share a cube?
 $a^3 = b^3 \Rightarrow a = b$ for real numbers — no.
- 3 So at most one arrow arrives anywhere.

ANSWER

One-to-one, and therefore a function

EXAMPLE 2 Is $x \mapsto \sqrt{x}$ with output “any square root of x ” a function?

① $x = 9$ maps to 3 and to -3 .

Two arrows leave one input.

② The definition demands exactly one.

ANSWER

No — it is one-to-many, not a function

EXAMPLE 3 The mapping “capital city of” on {Uzbekistan, France, Japan}. Classify it.

① Each country has exactly one capital.

A function.

② No two of these countries share a capital.

At most one arrow arrives.

ANSWER

One-to-one

03 Interactive model

Drag the inputs to see when a mapping stops being a function.

Function or not? $x \mapsto 2x - 7$ is: $x \mapsto |x|$ is:Which of these is **not** a function?

$x \mapsto 5$ for every x is:

not a function

one-to-one

many-to-one

one-to-many

A mapping can be undone by a mapping when it is:

many-to-one

one-to-one

one-to-many

any function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Mapping	Akslantirish	Отображение
Object (input)	Argument	Аргумент
Image (output)	Tasvir	Образ
One-to-one	Biror-birga	Взаимно однозначное
Many-to-one	Ko'pdan-birga	Многие в одно
One-to-many	Birdan-ko'pga	Одно во многие
Arrow diagram	Strelkali diagramma	Стрелочная диаграмма
Ordered pair	Tartiblangan juftlik	Упорядоченная пара
Function	Funksiya	Функция

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A function is a mapping in which every input has:

$x \mapsto x^2$ on the reals is:

Ordered pairs $\{(1,2), (2,4), (1,5)\}$ describe:

The set of all first coordinates is the:

range

domain

image

codomain

06

Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Classify $x \mapsto x + 4$

ANSWER one-to-one

2. Classify $x \mapsto 3$

ANSWER many-to-one

3. Classify $x \mapsto \pm x$

ANSWER one-to-many

4. Is $\{(1,1), (2,4), (3,9)\}$ a function?

ANSWER yes

5. Is $\{(1,1), (1,2)\}$ a function?

ANSWER no

6. Classify $x \mapsto x^4$ on the reals

ANSWER many-to-one

7. Name the domain of $\{(0,5), (2,7)\}$

ANSWER $\{0, 2\}$

Medium · the standard question

7 PROBLEMS

1. Classify $x \mapsto 2^x$

ANSWER one-to-one

2. Classify $x \mapsto \sin x$ on the reals

ANSWER many-to-one

3. Is $x^2 + y^2 = 9$ a function of x ?

ANSWER no

4. Give a mapping on $\{a, b, c\}$ that is many-to-one

ANSWER e.g. all three $\mapsto 1$

5. Write $x \mapsto 3x - 1$ on $\{0, 1, 2\}$ as ordered pairs

ANSWER $\{(0, -1), (1, 2), (2, 5)\}$

6. Classify “mother of” on a set of people

ANSWER many-to-one

7. Classify “sibling of”

ANSWER many-to-many

Hard · stretch and reason

7 PROBLEMS

1. Restrict $x \mapsto x^2$ to make it one-to-one

ANSWER $x \geq 0$ (or $x \leq 0$)

2. Explain why one-to-many mappings can never be functions

ANSWER One input would need two images

3. Is $x \mapsto \frac{1}{x}$ one-to-one on $x \neq 0$?

ANSWER yes

4. Find all k making $x \mapsto x^2 + kx$ one-to-one on $x \geq 0$

ANSWER $k \geq 0$

5. How many functions map $\{1, 2\}$ to $\{a, b\}$?

ANSWER 4

6. How many of those are one-to-one?

ANSWER 2

7. A mapping on a finite set is one-to-one exactly when it is what?

ANSWER onto — it uses every output

07 Homework

Homework — 4 tasks

Finish the project sheet if your pair did not.

1. Complete the four parts of the project sheet.
2. Classify: $x \mapsto x^5$, $x \mapsto x^6$, $x \mapsto |x| + 1$.
3. Write a mapping from your own life that is many-to-many.
4. Explain in two sentences why every one-to-one mapping is a function but not every function is one-to-one.